

Mostafa **Kermani Nia**

AI/ML Engineer | Applied AI & Data Engineering

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Profile

AI/ML Engineer with hands-on experience designing and deploying production AI systems, specializing in LLM-powered pipelines, computer vision, semantic retrieval, and data engineering. Built end-to-end solutions spanning data ingestion, intelligent information extraction, vector search, evaluation, and automation for real-world products. Currently conducting research in Byzantine-robust federated learning and vision-language models. Ranked 1st in Computer Engineering for eight consecutive semesters at the University of Tehran and serving as Head TA for undergraduate and graduate Machine Learning.

Education & Honors

University of Tehran — B.Sc. Computer Engineering — **GPA: 19.72/20**
Ranked 1st in cohort for 8 consecutive semesters — Sep 2022–Present

National University Entrance Examination — Top **0.6%** Nationwide

Iranian National Physics Olympiad — **Silver Medalist** (2021)

Industry Experience

ModAI — AI-powered fashion discovery and commerce

Tehran, Iran

Senior Applied AI & Data Engineer

- **Production AI & multimodal data pipelines:** Architected and implemented an end-to-end video-to-product intelligence system, processing merchant Instagram content through automated crawling, LLM-based SKU extraction, multi-product association, PostgreSQL matching, media ingestion, S3-compatible storage, and backend API integration. Designed persistent processing states, incremental crawling, duplicate-SKU handling, automated scheduling, and structured reporting; worked across historical crawls totaling roughly **50K posts and 5K video candidates** (not all matched or ingested).
- **Behavioral analytics & personalization:** Designing user- and device-level behavioral intelligence using **ClickHouse, PostgreSQL, and Elasticsearch**, integrating event streams, product interactions, search behavior, engagement signals, and purchase-conversion events into analytical profiles, actionable metrics, dashboards, and APIs.
- **LLM-powered data enrichment:** Developed structured product-description and catalog-enrichment pipelines involving taxonomy design, prompt engineering, schema validation, error analysis, and automated quality reporting to improve product understanding and downstream retrieval.
- **Software engineering & production optimization:** Contributed to large-scale commerce applications, including reducing Cumulative Layout Shift from approximately **1.2 to below 0.1** and delivering production mobile-commerce integrations.

RamonAI — intelligent commerce and visual search

Tehran, Iran

Applied AI Engineer

- **Computer vision & vector retrieval:** Designed and implemented a production visual-search pipeline combining dynamic screenshot-region extraction, image representations, and **Elasticsearch dense-vector retrieval** to match social-media content against real e-commerce product catalogs. Contributed to embedding optimization and semantic-search efficiency.
- **LLM evaluation & search intelligence:** Developing an LLM-based evaluation framework and API for classifying search results by relevance, detecting retrieval failures, and identifying unmet customer demand and potential commercial opportunities.
- **Data-driven product intelligence:** Analyzed production search, product, and business data across **PostgreSQL, ClickHouse, and Elasticsearch/Kibana**, translating raw records into measurable retrieval-quality indicators and decision-support analytics.
- **End-to-end engineering:** Integrated AI capabilities into production application workflows and backend APIs, connecting retrieval and model evaluation to commerce operations.

MOMIS Studio — social gaming and software engineering

Pomegroup | Finland / Remote

Co-Founder & Technical Lead

- **Product engineering & technical leadership:** Co-founded and led technical development of a multi-product software initiative spanning approximately **two years**, taking products from initial concepts through architecture, implementation, testing, and releases while coordinating a distributed development team.
- **Algorithmic systems & fraud detection:** Architected interactive game systems and implemented server-side heuristic anti-cheat mechanisms using gameplay timing, behavioral rules, and anomaly detection to identify suspicious activity and improve competitive integrity.
- **Scalable application infrastructure:** Developed Telegram-based applications, backend integrations, database-driven workflows, and Web3 reward mechanisms; established technical collaboration with the ONTON ecosystem.
- **Data-informed product development:** Used engagement patterns, user feedback, and community experiments to guide product iterations; organized tournaments reaching approximately **1,000 participants in one week**.
- **Engineering management:** Directed roadmap execution, technical planning, and stakeholder coordination across distributed contributors.

ONTON Platform — Web3 community engagement

Pomegroup | Finland / Remote

Backend Developer

- **Backend systems & blockchain integration:** Contributed to TON-based transaction verification, wallet authentication, payment processing, and reward distribution workflows, including handling transaction-related failures and improving application reliability.
- **Scalable event processing:** Worked on blockchain event-listening and application integration for a platform with a reported **900K+ users**, supporting data consistency across Web3-enabled user interactions.

Grantia — intelligent grant discovery and matching

Pikan Ventures AB | Sweden

AI Software Engineer & Technical Lead

- **RAG & recommendation systems:** Architected a multi-stage grant-matching system combining deterministic SQL eligibility filters, **pgvector semantic retrieval**, and **Gemini-based reasoning** to evaluate candidate matches and generate explainable, context-aware recommendations.
- **Data engineering & automation:** Built Python/n8n ETL workflows for collecting, cleaning, normalizing, and integrating heterogeneous European grant information into **PostgreSQL/Supabase**, with automated monitoring, execution logs, failure alerts, and operational reporting.
- **Technical ownership:** Led system architecture and implementation decisions, translating business requirements into production-oriented engineering workflows and coordinating technical delivery with product stakeholders.

Majara — AI-assisted event aggregation and management

Pikan Ventures AB | Sweden

AI Software Engineer & Technical Lead

- **LLM-powered data processing:** Engineered automated Python/n8n pipelines for web-data ingestion, LLM-assisted structured extraction, translation, cleaning, and normalization of heterogeneous event information.
- **Intelligent workflow orchestration:** Designed PostgreSQL-backed data workflows integrating generative-AI visual enrichment, scheduled publishing, validation, event-management operations, and automated error monitoring.
- **End-to-end system architecture:** Led development across data ingestion, application logic, external integrations, and operational observability, transforming manual business workflows into an integrated software platform.

Research Experience

FedMed-Defender: Byzantine-Robust Federated Learning *(ongoing)*

University of Tehran

Research collaborator — advisor: Dr. Mohammadreza Shokournia

- Developing **FedMed-Defender**, a **PyTorch** experimental framework for Byzantine-robust federated aggregation under **non-IID** medical data and malicious client updates; implementing and evaluating aggregation baselines and adversarial scenarios while investigating robust statistics, projection-space representations, and update filtering.

Vision-Language Models & Mechanistic Interpretability *(ongoing)*

University of Tehran

Research collaborator — advisor: Prof. Mostafa Tavassolipour

- Investigating **vision-language models (VLMs)** through exploratory multimodal experiments on how controlled image and question variations affect predictions and internal representations; refining hypotheses and evaluation workflows for model interpretability.

Selected AI & Data Projects

ORBIT — Local-First Voice AI Agent

Python, faster-whisper, Ollama/Qwen, tool calling, local memory

- Built an inspectable voice-agent stack with on-device ASR, LLM-driven tool selection, speech synthesis, local memory, desktop automation, telemetry, and safe-action controls; includes tests and architecture documentation.

Speech Processing ML Toolkit

Python, librosa, scikit-learn, TensorFlow

- Designed a reusable speaker-identification/gender-classification workflow with signal cleaning, MFCC and prosodic feature extraction, and SVM/KNN/ensemble/MLP experiments and evaluation.

Mathematics Misconception Analytics Pipeline

Python, pandas, SQL, TF-IDF, Docker, GitHub Actions

- Engineered a reproducible CSV-to-MySQL **ETL/feature-engineering pipeline** for educational text, with normalization, categorical features, TF-IDF and CI/container deployment configuration.

Computer Vision & Deep Learning Experiments

Python, deep learning, neural network initialization, model comparison

- Implemented and compared pretrained CNN and generative-model experiments; independently studied **Xavier/Glorot versus Kaiming/He initialization** and **optimization/imbalanced-learning** settings in reproducible notebooks.

Technical Skills

ML & Statistics	PyTorch, TensorFlow, scikit-learn, feature engineering, supervised/unsupervised learning, SVM, ensembles, model evaluation, Computer Vision, speech/audio ML, federated learning (research)
GenAI & Retrieval	LLM integration, RAG, prompting/evaluation, multimodal processing, embeddings, vector similarity, Gemini/OpenAI APIs, Whisper, Ollama, agent/tool workflows
Data & Platforms	Python, SQL, PostgreSQL, ClickHouse, Elasticsearch, pgvector, Kafka (working familiarity), pandas, ETL/ELT, data quality, Metabase, Kibana
Engineering	REST APIs, Docker, Git/GitHub Actions, n8n, Linux, S3-compatible object storage, SQLite, Next.js/TypeScript (supporting skills)

Academic Teaching & Leadership

Head Teaching Assistant — Machine Learning (B.Sc. & M.Sc.) *Tehran, Iran*

University of Tehran

- Lead TA coordination, assignment design, project mentoring, and assessment across undergraduate and graduate Machine Learning courses, covering statistical learning, optimization, and model evaluation. *Course instructors:* **Prof. Babak Nadjar Araabi**, **Prof. Mostafa Tavassolipour**, and **Prof. M. R. Abolghasemi**.

Course Instructor — Speech & Audio AI *Iran / Remote*

Maktabkhooneh — Online Learning Platform

- Designed a comprehensive **six-chapter, project-based curriculum**, planned for **35+ lessons and approximately 25–30 hours** of instruction spanning digital signal processing, classical ML, deep learning, and generative speech systems.
- Developed practical Python projects covering **FFT/STFT, MFCC, speech classification, Whisper ASR, speaker verification, TTS/XTTS, voice cloning**, and an end-to-end LLM-powered voice assistant.

Head Teaching Assistant — Data Science *Tehran, Iran*

University of Tehran

- Co-led **20+ TAs** supporting **100+ students**; contributed to curriculum development, practical assignments, and evaluation across statistical learning, ML/DL, SQL, LLMs, RAG, AI agents, and Big Data/MLOps. *Course instructors:* **Prof. Yadollah Yaghoobzadeh** and **Dr. S. Fatemeh Razavi**.

Head Teaching Assistant — Probability & Statistics *Tehran, Iran*

University of Tehran

- Led **33 TAs** supporting **180+ students**; coordinated assignments, assessment, and academic operations covering probability theory, random variables, statistical inference, and applied statistics. *Course instructors:* **Dr. Behnam Bahrak** and **Dr. Abdolhossein Vahabie**.

Teaching Assistant — Artificial Intelligence *Tehran, Iran*

University of Tehran

- Designed and supported assignments in classical search algorithms, constraint satisfaction problems, and machine-learning fundamentals. *Course instructor:* **Dr. Hakimeh Fadaei**.

Teaching Assistant — Natural Language Processing *Tehran, Iran*

K. N. Toosi University of Technology

- Support undergraduate NLP instruction and practical coursework covering text processing, linguistic representations, and machine-learning approaches to natural language understanding.

Teaching Assistant — Computer Vision *Tehran, Iran*

University of Tehran

- Supported theoretical and programming assignments in image processing, feature extraction, visual recognition, and fundamental computer-vision methods.

Teaching Assistant — Speech & Audio Signal Processing *Hamedan, Iran*

Bu-Ali Sina University

- Designed and recorded project-based supplementary instruction covering digital signal processing, audio feature extraction, speech classification, ASR, TTS, and speaker verification using Python. *Course instructor:* **Dr. Mahdi Seifipour**.

Teaching Assistant — Data Science & Simulation *Tehran, Iran*

Tehran University of Medical Sciences

- Support doctoral-level coursework in data fundamentals and computational simulation for allied-health and radiology students. *Course instructor:* **Dr. Mahdi Seifipour**.

Languages

Persian: Native · **English:** Professional working proficiency